

KoalaFalcon: Lattice Signature Aggregation with SNARKs over KoalaBear (Preliminary Draft)

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Abstract

SNARK-based lattice signature aggregation faces the challenge of non-native arithmetization to encode ring-arithmetic at the coefficient level over SNARK typical fields. In this work, we present KoalaFalcon, a variant of the Falcon signature scheme over the KoalaBear prime that uses Poseidon for message hashes. We present two parameter sets for KoalaFalcon achieving a UF-CMA bit security of 117.98 and 256 respectively using the techniques of CoreFalcon+, arguing the prime modulus does not affect the applicability of the security reduction so long as the intended ring-algebraic properties are met, and the signature sampler standard deviation is above a lattice specific smoothing parameter. Adherence to these criteria results in instances with concretely hard estimates for SIS based on the core-SVP methodology.

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1 Introduction

SNARK-based signature aggregation refers to proving knowledge of the correct verification of many signatures in a single SNARK proof. Given the quantum threat, it is desirable to update the massively deployed elliptic curve based cryptosystems to quantum resistant alternatives. Despite

post-quantum solutions to signatures and SNARKs both being available, there is still debate as to how best to scalably aggregate signatures in Ethereum.

Falcon has emerged as a promising post-quantum alternative to existing elliptic curve signatures. Falcon is lattice-based and prized for its compactness and speed. Though, its verification algorithm introduces multiple challenges regarding the establishment of a computational representation amenable to arithmetization for SNARKs. Firstly, Falcon signature arithmetic relies upon polynomial rings with coefficients modulo $q = 12289$. The field $\mathbb{F}_q$ is significantly smaller than those typical in modern hash-based SNARKs (e.g. KoalaBear). Encoding $\mathbb{F}_q$ arithmetic within the KoalaBear prime field requires non-native arithmetic, a process in which foreign field arithmetic is simulated in the native field. Doing so adds extra constraints for every foreign field operation compared to operations over the native field. As well, the size of $\mathbb{F}_q$ is less than 14 bits, too small to permit Reed-Solomon encodings of the moderate witness size 2^{14} . As a result, encoding and commitment would need to be done over an extension field, a performance degradation for polynomial commitments based on FRI. Furthermore, Falcon signature verification also employs arithmetic over the integers for ℓ_2 norm checks. As a result, the SNARK must be capable of handling both integer and cyclotomic ring arithmetic in addition to any non-native arithmetic.

In this work we present KoalaFalcon, a variant of Falcon modified to work over the KoalaBear prime modulus. Our primary modifications are to fix modulus q to the KoalaBear prime, and swap the hash function to Poseidon. This approach enables the ring arithmetic component of KoalaFalcon signature verification to be arithmetized at the coefficient level without non-native arithmetic for SNARKs defined over the KoalaBear prime field. As a result, SNARK proofs of aggregate KoalaFalcon signatures may rely on much of the same cryptographic infrastructure as existing hash-based SNARKs (e.g. LeanVM), making them more easily incorporated into recursive hash-based SNARKs.

We analyze how our modifications to Falcon affect the security of the resulting signature scheme, arguing that our changes are amenable to the same style of argument employed by CoreFalcon+, and retain similar security to the original Falcon scheme. The reasoning behind this position argues that changing the modulus in CoreFalcon+ does not impose new consequences to the applicability of the Rényi divergence based approach to the security reduction from CoreFalcon+ forgery to t - $\mathcal{R}$ -ISIS under the standard CoreFalcon+ approach to parameterization. We define two parameter sets for KoalaFalcon achieving UF-CMA bit security estimates of 117.98 and 256 respectively using these techniques. As well, we observe that our estimates remain similar to the original Falcon scheme despite these changes, arguing why by analyzing the impact of the parameter change on lattice reduction attacks.

1.1 Related Work

SNARKs SNARK proof systems prove and verify computational claims by combining an interactive oracle proof (IOP) with a polynomial commitment scheme (PCS). They are made non-interactive by simulating verifier responses with the Fiat-Shamir heuristic. In general, the design choices behind a SNARK proof system can be roughly divided into those about *arithmetization* and those about the *PCS*. Considerations of arithmetization revolve around how applications are compiled into arithmetic circuits, and what IOPs are used to verify that a witness attests to the correct execution of a computation. The PCS necessarily corresponds to how arithmetization is performed, as the prover must make a binding commitment to the witness of an arithmetized computation prior to any interaction. The commitment is later opened at a point determined over the course of the IOP. The field of arithmetization must be compatible enough with the PCS that this opening can be made in order for the soundness of the IOP to be established with respect to the

initial commitment. Non-native arithmetic is commonly used when applications are defined over fields incompatible with the chosen field of arithmetization (a process where arithmetic for a foreign field is simulated within the field of arithmetization).

Arithmetic Hashes and Recursive SNARKs Arithmetic hashes are hash functions designed specifically to be efficiently incorporated into arithmetic circuits. The best example being Poseidon [16], whose permutation function consists entirely of arithmetic operations. Poseidon has become a standard primitive in the construction of Recursive SNARKs, which refer to SNARK proof systems that partition the full execution trace of computations too large to feasibly prove within a single proof, recursively proving those partitions. Proof recursion is achieved by starting with a proof of the first partition, then in each recursive proof simultaneously proves the next partition while verifying the previous in the same circuit.

Primitive-Friendly SNARKs vs SNARK-Friendly Primitives SNARKs which prove the execution of cryptographic primitives that are not easily arithmetized often utilize *SNARK-Friendly* primitives, such as Poseidon. An alternative approach to the non-native arithmetic problem is to use a *Primitive-Friendly* SNARK. Examples include Binius [9], Flock [7], and Labradoor [4]. Such schemes define arithmetization schemes that efficiently encode applications with special forms of arithmetic. Binius and Flock, for example, are especially efficient at proving the execution of binary circuits.

A downside of primitive-friendly SNARKs is their bespoke nature. When the proof system is designed around a single arithmetization objective, it may become difficult to compose such proofs into larger proof systems (e.g. recursive SNARKs) without the reintroduction of non-native arithmetic. For example, if a primitive-friendly SNARK uses arithmetic incompatible with that of a SNARK-friendly hash, it is not straightforward how to combine them into a unified protocol. As well, if the PCS of the primitive-friendly SNARK uses arithmetic incompatible with that of the broader system, the task of batching PCS proofs is also no longer straightforward. Such incompatibilities introduce additional design and security considerations beyond those standard to composable proofs.

SNARK-Based XMSS Signature Aggregation Another approach to post-quantum signature aggregation is SNARK-based aggregation of XMSS signatures [6]. The XMSS approach to signatures involves taking a one-time signature and making it into a many-time signature by creating a Merkle root of their public keys. An inclusion proof alongside a one-time signature becomes a many-time signature, with aggregate verification provable with hash-based SNARKs [11]. SNARK-based aggregation of XMSS signatures suffer from large circuits dominated by hashing. They are also stateful, requiring more considerations placed on the signer to remain secure in practice.

Falcon Signature Aggregation with Labradoor Labradoor [4] is a lattice-based SNARK for proving R1CS. Recently, it has been used to prove Falcon signatures in aggregate [1] and serves as the most direct performance comparison to the approach taken in this work.

1.2 Contributions

- A variant of CoreFalcon+ (KoalaFalcon) over the KoalaBear prime modulus using Poseidon for message hashes. We demonstrate the existence of parameters that retain provable security under the Rényi divergence approach to provable security. These parameters result in favorable hardness estimates via the core-SVP methodology, achieving a UF-CMA bit security of

117.98 and 256 bits for $\lambda = 128$ and $\lambda = 256$ respectively. We argue that aside from two conditions on the algebraic structure implied by concrete parameter choices, and an adherence to a lower bound on the standard deviation for signature sampling, the reduction applies for general moduli.

2 Preliminaries

2.1 Notation

Let n be a power-of-two, q is prime, $R = \mathbb{Z}[X]/(X^n + 1)$, and $R_q = R/qR$. In the context of Falcon, we will refer to $\alpha \in \mathbb{R}^{>1}$ as the trapdoor basis quality, $\beta \in \mathbb{R}^{>0}$ as the signature norm bound, $k \in \mathbb{N}$ as the salt bit length, $s \in \mathbb{R}^{>0}$ as the signature Gaussian standard deviation, tailcut rate $\tau > 1$, $\epsilon > 0$, security parameter λ , max number of signing queries $Q_s \in \mathbb{N}$, max number of random oracle queries $Q_H \in \mathbb{N}$, max number of signing attempts $C_s \in \mathbb{N}$, and Rényi orders a_u and a_p . For a lattice basis B , write $\|B\|_{\text{GS}}$ for its Gram–Schmidt norm. For $x \in \mathbb{R}^m$, write $\|x\|_2$ for its ℓ_2 norm. Write $U(R_q)$ for the uniform distribution on R_q . Write $D_{\Lambda, s}$ for the discrete Gaussian on a lattice Λ with width s . Write $\eta_\epsilon(\Lambda)$ for the smoothing parameter of Λ at slack ϵ . Write $R_a(P \| Q)$ for Rényi divergence of order a between distributions P and Q . We will sometimes refer to elements of R, R_q as matrices in reference to their (-1) -skew-circulant representation, which preserves the commutativity of cyclotomic ring multiplication.

2.2 Definitions

Definition 2.1 (Theorem 2.9 [17]). *Let $\Lambda = \Lambda(B_1)$ be a lattice with basis B_1 . A matrix B_2 is also a basis of Λ if and only if there exists a unimodular matrix U with $\det U = \pm 1$ such that $UB_2 = B_1$.*

Definition 2.2 (Regev, §3 [18]). *Let q be prime, $A \in \mathbb{Z}_q^{n \times m}$ have linearly independent rows, and $m \geq n$. The m -dimensional q -ary orthogonal lattice associated with A is*

$$\Lambda_q^\perp(A) = \{y \in \mathbb{Z}^m : Ay = 0 \pmod{q}\}.$$

Its determinant is $\det(\Lambda_q^\perp(A)) = q^n$.

Definition 2.3 (Definition 2.11 [17]). *The volume of a lattice is its determinant and measures the volume of its fundamental parallelepiped.*

Definition 2.4 (Definition 2.12 [17]). *The i -th successive minimum of an m dimensional lattice $\lambda_i \in \{\lambda_1, \dots, \lambda_m\}$ is the smallest real number r such that the lattice contains i linearly independent vectors at most length r . Intuitively, each λ_i is the smallest radius of an m dimensional ball centered at the origin such that it contains each of these i linearly independent vectors of the lattice.*

Definition 2.5 (Regev, §3 [18]; [8]). *For $\Lambda_q^\perp(A)$,*

$$\lambda_1(\Lambda_q^\perp(A)) \approx q^{n/m} \sqrt{\frac{m}{2\pi e}}.$$

Definition 2.6 (Definition 10 [12]; Regev, §3 [18]). *For $h \in R_q \setminus \{0\}$ and $c \in R_q$, define*

$$\Lambda_h = \{(u, v) \in R^2 : u + vh = 0 \pmod{q}\}, \quad \Lambda_{h,c} = \{(u, v) \in R^2 : u + vh = c \pmod{q}\}.$$

Then $\Lambda_{h,c} = \Lambda_h + t$ for some shift t is called the shifted lattice (or lattice coset) of Λ_h , and corresponds to a translation by t . With $m = 2n$,

$$\lambda_1(\Lambda_h) \approx q^{n/m} \sqrt{\frac{m}{2\pi e}} = \sqrt{\frac{qn}{\pi e}}.$$

Definition 2.7 (Definition 10 [12]). *In the (Q_H+1) -R-ISIS game, the adversary receives $(h, c_1, \dots, c_t)$ with $h \leftarrow \text{TpGen}(\alpha, q)$, each $c_i \leftarrow U(R_q)$, and $t = Q_H + 1$. It must output a short pair $(u, v) \in R^2$ and an index j such that $c_j = u + vh \pmod{q}$ and $\|(u, v)\|_2 \leq B$.*

Definition 2.8 (Lemma 3.1 [15]). *For any n -dimensional lattice Λ and real $\epsilon > 0$*

$$\eta_\epsilon(\Lambda_{h,q}) \leq \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \|B\|_{GS, \min}.$$

Where $\|B\|_{GS, \min}$ is the Gram-Schmidt minimum across all bases B of Λ .

Definition 2.9 (Theorem 2.9 [17]; [12]). *Let q be prime, $n = 2^\kappa$, $\phi = X^n + 1$ split over $\mathbb{Z}_q$, and $f, g \in R_q^\times$ with*

$$h \equiv gf^{-1} \pmod{\phi, q}, \quad fG - gF \equiv q \pmod{\phi}.$$

The private and public bases of the NTRU lattice are

$$B_1 = \begin{bmatrix} g & -f \\ G & -F \end{bmatrix}, \quad B_2 = \begin{bmatrix} -h & I_n \\ qI_n & O_n \end{bmatrix},$$

with ring elements represented as (-1) -skew-circulant matrices.

Definition 2.10 (Lemma 4 [12]; Lemma 3.1 [15]). *CoreFalcon+ sets*

$$\eta_\epsilon^{\text{upper}}(\mathbb{Z}^{2n}) = \frac{1}{\pi} \sqrt{\frac{\ln\left(4n\left(1 + \frac{1}{\epsilon}\right)\right)}{2}}, \quad s = \eta_\epsilon^{\text{upper}}(\mathbb{Z}^{2n}) \cdot \alpha\sqrt{q}, \quad \beta = \tau s \sqrt{2n}.$$

Trapdoor generation satisfies $\|B\|_{GS} \leq \alpha\sqrt{q}$. The smoothing condition used by the reduction is $s \geq \eta_\epsilon(\Lambda_{h,q})$; GPV, Lem. 3.1 [15] gives $s \geq \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \|B_{f,g}\|_{GS}$.

Corollary 2.11 (Corollary 2 [12]). *Let q be prime, $h \in R_q \setminus \{0\}$, $a \in (1, \infty)$, $\epsilon \in (0, \frac{1}{2})$, $s \geq \eta_\epsilon(\Lambda_{h,q})$, $P = U(R_q)$, and Q the distribution of $u + v \cdot h \pmod{q}$ with $u, v \sim D_{R,s}$. Then*

$$R_a(P \| Q) \lesssim 1 + \frac{2a\epsilon^2}{(1 - \epsilon)^2}.$$

Corollary 2.12 (Corollary 1 [12]). *Let n be a positive integer, $a \in (1, \infty)$, $\epsilon \in (0, 1/4)$. Then for $\text{PreSmp} := \text{FFO Sampler}$, and lattice $\Lambda = \Lambda(B)_{(c,0)}$ for any basis $B \in \mathbb{Z}^{2n \times 2n}$, $s \geq \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \|B\|_{GS}$, and arbitrary syndrome $c \in R_q$, the Rényi divergence is bounded by*

$$R_a(\text{PreSmp}(B, s, (c, 0)) \| D_{\Lambda, s}) \lesssim 1 + 2a\epsilon^2.$$

Lemma 2.13 (Lemma 7 [12]). *Let $n \in \mathbb{Z}^+$, $\epsilon \in (0, 1/4)$. Then the relative error of $\text{PreSmp} := \text{FFO Sampler}$ and lattice $\Lambda = \Lambda(B)$ for any basis $B \in \mathbb{Z}^{2n \times 2n}$, $s \geq \eta_\epsilon(\mathbb{Z}^{2n \times 2n}) \cdot \|B\|_{GS}$, and arbitrary syndrome $c \in R_q$ is bounded by*

$$\delta_{\text{RE}}(\text{PreSmp}(B, s, (c, 0)), D_{\Lambda, s}) \leq \left(\frac{1 + \epsilon/(2n)}{1 - \epsilon/(2n)} \right)^{2n} - 1 \approx 2\epsilon$$

Lemma 2.14 (Lemma 10 [12]). *Let P, Q be two distributions sharing the same space of outputs with non-zero probability $\sup(Q) = \sup(P) = \mathbb{Z}^n$ and $\delta_{RE}(P, Q) = \delta$. Then for any $\beta \geq 0$*

$$\Pr_{x \leftarrow P}[\|x\|_2 > \beta] \leq \Pr_{x \leftarrow Q}[\|x\|_2 > \beta] \cdot (1 + \delta)$$

Lemma 2.15 (Lemma 2 [12]). *For any lattice $\Lambda \subseteq \mathbb{R}^n$, width $s > 0$, shift $t \in \mathbb{R}^n$, and tailcut rate $\tau > 1$*

$$\Pr_{z \leftarrow D_{\Lambda+t, s}}[\|z\|_2 > \tau s \sqrt{n}] \leq \frac{\rho_s(\Lambda)}{\rho_s(\Lambda + t)} \left(\sqrt{e^{1-\tau^2} \tau^2} \right)^n$$

Lemma 2.16 (Lemma 5 [12]). *For any n -dimensional lattice Λ , center $c \in \mathbb{R}^n$, and reals $0 < \epsilon < 1$, and $s \geq \eta_\epsilon(\Lambda)$, we have*

$$\rho_{s,c}(\Lambda) \in [1 - \epsilon, 1 + \epsilon] \cdot \frac{(\sqrt{2\pi} \cdot s)^n}{\det(\Lambda)}$$

Definition 2.17 (Theorem 1 [12]).

$$\begin{aligned} \text{Adv}_{\text{CoreFalcon}^+, A}^{Q_s\text{-UF-CMA}} &\leq \left(r_u^{C_s} \cdot \left(r_p^{C_s} \cdot \text{Adv}_{q, \alpha, \beta, B}^{(Q_H+1)\text{-R-ISIS}} \right)^{\frac{a_p-1}{a_p}} \right)^{\frac{a_u-1}{a_u}} \\ &\quad + \sum_{i=0}^{Q_s} \binom{C_s}{i} (1 - p_{\text{PreSmp}, \beta})^{C_s-i} (p_{\text{PreSmp}, \beta})^i + \frac{C_s(Q_H + C_s)}{2^k} \end{aligned}$$

$$p_{\text{PreSmp}, \beta} := \min_{(B, \cdot) \in \text{TpDGen}, c \in R_q} \Pr_{(s_1, s_2) \leftarrow \text{PreSmp}(B, s, (c, 0))}[\|(s_1, s_2)\|_2 \leq \beta],$$

$$r_u = \max_{(\cdot, h) \in \text{TpDGen}} R_{a_u}(P \| Q_h),$$

$$r_p = \max_{(B, \cdot) \in \text{TpDGen}, c \in R_q} R_{a_p}(\text{PreSmp}(B, s, (c, 0)) \| D_{\Lambda, s}),$$

Where $\text{Adv}_{q, \alpha, \beta, B}^{(Q_H+1)\text{-R-ISIS}}$ is the advantage adversary A has in determining a pre-image $(u, v) \in R^2$ satisfying $c_j = u + vh \pmod q \wedge \|(u, v)\|_2 \leq B$ where $B > 0$ is the R-ISIS solution norm bound, $c_j \in \{c_1, \dots, c_t\}$ are uniformly sampled. $p_{\text{PreSmp}, \beta}$ is the worst-case probability of the FFO Sampler algorithm correctly sampling an $(s_1, s_2) \in R^2$ such that $\|(s_1, s_2)\|_2 \leq \beta$. r_u is the multiplicative Rényi divergence $R_{a_u}(P \| Q_h) \geq 1$ (order a_u) between $P = U(R_q)$ and Q_h , the distribution of $s_1 + s_2 \cdot h \pmod q$ for $s_1, s_2 \sim D_{R, s}$. Values $r_u \approx 1$ indicate negligible loss, consistent with the factor $r_u^{C_s}$ in the bound. r_p is the multiplicative Rényi divergence $R_{a_p}(\text{PreSmp}(B, s, (c, 0)) \| D_{\Lambda, s}) \geq 1$ (order a_p) between FFO sampler output and the discrete Gaussian $D_{\Lambda, s}$, where $\Lambda = \Lambda(B)_{(c, 0)}$ is the shifted preimage coset for syndrome $(c, 0)$; $r_p \approx 1$ is the good case, consistent with $r_p^{C_s}$. C_s is the max number of attempts to sample (s_1, s_2) a valid signature satisfying $\|(s_1, s_2)\|_2 \leq \beta$.

Definition 2.18 (Regev, §3 [18]; [14]). *For a random m -dimensional q -ary lattice $\Lambda_q^\perp(A)$, lattice reduction with root-Hermite factor δ finds vectors of length approximately*

$$\min\{q, q^{n/m} \delta^m\}.$$

Typical values are $\delta \approx 1.013$ for fast BKZ and $\delta \approx 1.011$ for slower, more precise variants.

3 KoalaFalcon

3.1 Core Construction

KoalaFalcon is a variant of CoreFalcon+ (itself a variant of Falcon with a slight modification to better enable security analysis with Rényi divergence). The primary algorithmic difference in KoalaFalcon is to use Poseidon as the hash function instead of SHAKE. Beyond this, KoalaFalcon fixes the modulus q to the KoalaBear prime. To support the KoalaBear prime, we set the following formulaic parameters from CoreFalcon+, explaining the rationale and applicability of this choice in 3.2:

$$s = \frac{1}{\pi} \sqrt{\frac{\ln\left(4n \left(1 + \frac{1}{\epsilon}\right)\right)}{2}} \cdot \alpha \sqrt{q}, \quad \beta = \tau s \sqrt{2n}.$$

KoalaFalcon’s signing and verification algorithms are depicted in Figure 1, with two concrete parameter sets defined for KoalaFalcon in Table 1. Algorithmically, the primary difference between CoreFalcon+ and KoalaFalcon is in the hash function. KoalaFalcon is able to use Poseidon natively because it fixes q as the KoalaBear prime, supporting pre-existing Poseidon instantiations under the same modulus. Messages are hashed into R_q elements by taking repeated Poseidon permutations, collecting the sponge state between each invocation until n coefficients have been collected. The rest remains algorithmically unchanged in KoalaFalcon aside from re-parameterization.

Gen 01 $(f, g, F, G, h) \xleftarrow{\$} \text{TpGen}(\alpha, q)$ 02 $B := \begin{bmatrix} g & -f \\ G & -F \end{bmatrix} \in \mathbb{F}_q^{2n \times 2n}$ 03 return $(sk := B, pk := h)$	
COREFALCON⁺ Sgn⁺ ($sk = B, m$) 01 repeat 02 $r \xleftarrow{\$} \{0, 1\}^k$ 03 $c := H(pk, r, m) \in R_q$ 04 $(s_1, s_2) \xleftarrow{\$} \text{PreSmp}(B, s, (c, 0))$ 05 until $\ (s_1, s_2)\ _2 \leq \beta$ 06 $\sigma := (r, s_2) \in \{0, 1\}^k \times R$ 07 return σ Ver ($pk = h, m, \sigma = (r, s_2)$) 08 $c := H(pk, r, m)$ 09 $s_1 := c - s_2 \cdot h \pmod{q}$ 10 return $\mathbb{I}[\ (s_1, s_2)\ _2 \leq \beta]$	KOALAFALCON KoalaSgn ($sk = B, m$) 01 repeat 02 $r \xleftarrow{\$} \{0, 1\}^k$ 03 $c := \text{Poseidon}(pk, r, m) \in R_q$ 04 $(s_1, s_2) \xleftarrow{\$} \text{PreSmp}(B, s, (c, 0))$ 05 until $\ (s_1, s_2)\ _2 \leq \beta$ 06 $\sigma := (r, s_2) \in \{0, 1\}^k \times R$ 07 return σ KoalaVer ($pk = h, m, \sigma = (r, s_2)$) 08 $c := \text{Poseidon}(pk, r, m)$ 09 $s_1 := c - s_2 \cdot h \pmod{q}$ 10 return $\mathbb{I}[\ (s_1, s_2)\ _2 \leq \beta]$

Figure 1: KoalaFalcon and CoreFalcon⁺ key generation, signing, and verification. KOALAFALCON = (Gen, KoalaSgn, KoalaVer) and COREFALCON⁺ = (Gen, Sgn⁺, Ver).

KOALAFALCON-512	KOALAFALCON-1024
$\lambda = 128$	$\lambda = 256$
$n = 512$	$n = 1024$
$q = 2^{31} - 2^{24} + 1$	$q = 2^{31} - 2^{24} + 1$
$Q_s = 2^{64}$	$Q_s = 2^{64}$
$C_s = 2^{64} + 2^{50}$	$C_s = 2^{64} + 2^{50}$
$Q_H = 2^{96}$	$Q_H = 2^{96}$
$\alpha = 1.17$	$\alpha = 1.17$
$k = 416$	$k = 416$
$\tau = 1.1$	$\tau = 1.1$
$\epsilon = 1/\sqrt{Q_s \cdot \lambda}$	$\epsilon = 1/\sqrt{Q_s \cdot \lambda}$
$\beta = \tau s \sqrt{2n}$	$\beta = \tau s \sqrt{2n}$
$s = \frac{1}{\pi} \sqrt{\frac{\ln(4n(1+\frac{1}{\epsilon}))}{2}} \cdot \alpha \sqrt{q}$	$s = \frac{1}{\pi} \sqrt{\frac{\ln(4n(1+\frac{1}{\epsilon}))}{2}} \cdot \alpha \sqrt{q}$

Table 1: Concrete parameters for KoalaFalcon-512 and KoalaFalcon-1024.

3.2 Security Reduction

In this section we discuss implications to the CoreFalcon+ security analysis from setting modulus q to the KoalaBear prime.

Requirements on q for the Algebraic Feasibility of the Construction First, we discuss whether the KoalaFalcon construction in Figure 1 is algebraically feasible to instantiate with the KoalaBear prime $q = 2^{31} - 2^{24} + 1$. This involves ensuring the necessary ring-algebraic requirements are in place to support efficient ring multiplication and that the desired NTRU lattice structure exists to support trapdoor sampling. Fast ring multiplication in Falcon uses the NTT, requiring q to permit a primitive $2n$ -th root of unity in $\mathbb{Z}_q$. This is to enable $X^n + 1$ to split fully into n linear factors over $\mathbb{Z}_q$. As well, the public and private bases output from trapdoor sampling $B_h, B_{f,g}$ must remain bases of the same NTRU lattice. The properties hold $q = 2^{31} - 2^{24} + 1$ and are proven in Appendix A.1.

Requirements on q in the t - $\mathcal{R}$ -ISIS Game Next, we discuss whether the CoreFalcon+ security reduction requires any properties of q beyond the desired ring setting. At this stage, we are not concerned with hardness estimates, only whether the hardness estimates of $\text{Adv}_{q,\alpha,B,\mathcal{A}}^{t\text{-}\mathcal{R}\text{-ISIS}}$ (the advantage of an adversary in solving the t - $\mathcal{R}$ -ISIS game) are still applicable without modification to the reduction for $q = 2^{31} - 2^{24} + 1$.

$$\text{Adv}_{q,\alpha,B,\mathcal{A}}^{t\text{-}\mathcal{R}\text{-ISIS}} := \Pr[t\text{-}\mathcal{R}\text{-ISIS}_{q,\alpha,B}(\mathcal{A}) \Rightarrow 1].$$

The CoreFalcon+ security reduction demonstrates adversaries capable of forgery can be reduced to solvers of the t -R-ISIS problem depicted in Figure 2. The reduction is grounded in the one-wayness of the public-to-private key map and standard GPV proof strategies. Rather than arguing the applicability of the reduction to concrete instantiations via collision resistance and statistical distance of public keys to uniformity (as is the case for NTRUSign), CoreFalcon+ uses Rényi divergence to quantify the bit loss in security from the real signing and random oracle instantiation outputs to the ideal simulated setting. It is assumed t -R-ISIS is as hard as standard SIS and hardness estimates are informed from best-known SIS attacks via the core-SVP methodology.

A hybrid argument upper bounds a more directly applicable NTRU-ISIS advantage into an R-ISIS hardness plus decisional-NTRU advantage, arguing that t -R-ISIS is sufficient for the concrete security bounds on unforgeability. Rényi divergence is introduced into the bound as a multiplicative factor in the computational term to quantify the bit loss in security from simulated to real oracles.

Game t-R-ISIS$_{q,\alpha,B}(A)$ 01 $(\cdot, \cdot, \cdot, \cdot, h) \xleftarrow{\\$} \text{TpGen}(\alpha, q)$ 02 for $i \in [t]$ 03 $c_i \xleftarrow{\\$} R_q$ 04 $(j, u, v) \xleftarrow{\\$} A(h, c_1, \dots, c_t)$ 05 return $\llbracket u + h \cdot v = c_j \wedge \ (u, v)\ _2 \leq B \rrbracket$
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Figure 2: t -R-ISIS $_{q,\alpha,B}$ Game.

The t -R-ISIS game itself does not impose any strict requirement on q aside from the ring R_q existing. Thus, in the next section we turn our attention to how this ideal simulation in the ROM is bridged to a real instantiation, tracking exactly where the modulus enters into the security bound, and what requirements it must satisfy, term by term.

Requirements on q for the Security Reduction to t -R-ISIS The reduction from CoreFalcon+ forgery to a t -R-ISIS introduces Rényi-loss terms to quantify distributional differences between the idealized security game and a real instantiation. We argue that once the desired ring setting is established, the only q dependent property needed to maintain applicability of the CoreFalcon+ security bound under different moduli is that the signature standard deviation must be greater than or equal to the scaled smoothing parameter of the underlying lattice. While there are q dependent properties, the CoreFalcon+ formulaic parameters for ℓ_2 norm threshold β , target Gram-Schmidt norm of the private basis $\|B_{f,g}\|_{GS}$, and signature standard deviation s are set in such a way that this condition is fulfilled for generic q .

Lemma 3.1. *Fix $n, \epsilon, \alpha, \tau$. Let $\text{TpGen}(\alpha, q)$ return trapdoor basis $B_{f,g}$ with $\|B_{f,g}\|_{GS} \leq \alpha\sqrt{q}$. Where s, β are set as*

$$s = \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \alpha\sqrt{q}, \quad \eta_\epsilon(\mathbb{Z}^{2n}) = \frac{1}{\pi} \sqrt{\frac{\ln\left(4n\left(1 + \frac{1}{\epsilon}\right)\right)}{2}}, \quad \beta = \tau s \sqrt{2n}$$

Whether the smoothing condition $s \geq \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \|B_{f,g}\|_{GS}$ (and hence $s \geq \eta_\epsilon(\Lambda_{h,q})$) holds is independent of q for every modulus q in which R_q is defined.

Proof. Notice that $s \propto \sqrt{q}$ and $\beta \propto \sqrt{q}$, for fixed $n, \epsilon, \alpha, \tau$. In particular, when normalized by a factor of $\frac{1}{\sqrt{q}}$, the value of $\frac{s}{\sqrt{q}}, \frac{\beta}{\sqrt{q}}$ are both independent of q .

$$\frac{s}{\sqrt{q}} = \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \alpha$$

$$\frac{\beta}{\sqrt{q}} = \frac{\tau s \sqrt{2n}}{\sqrt{q}} = \tau \sqrt{2n} \cdot \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \alpha$$

The trapdoor sampler TpGen determines ring elements f, g, F, G defining $B_{f,g}$ satisfying

$$B_{f,g} = \begin{bmatrix} g & -f \\ G & -F \end{bmatrix}, \quad \|B_{f,g}\|_{GS} \leq \alpha\sqrt{q}.$$

The intended smoothing condition on s is $s \geq \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \|B\|_{GS}$. Since both s and the upper bound on $\eta_\epsilon(\mathbb{Z}^{2n}) \cdot \|B\|_{GS}$ scale proportional to $\sqrt{q}$, changing q preserves the inequality as long as the trapdoor-quality guarantee is maintained.

$$\eta_\epsilon(\mathbb{Z}^{2n})\|B_{f,g}\|_{GS} \leq \eta_\epsilon(\mathbb{Z}^{2n})\alpha\sqrt{q} = s.$$

Thus, for fixed (n, ϵ, α) , satisfaction of the smoothing condition is independent of q , provided the trapdoor-quality bound

$$\|B_{f,g}\|_{GS} \leq \alpha\sqrt{q}$$

continues to hold. □

Theorem 3.2. *Under the q -dependence audit of the CoreFalcon+ reduction given in [Appendix A](#), changing the modulus preserves the conditions required for the Rényi-loss analysis, provided the required cyclotomic ring setting and trapdoor-quality/smoothing conditions remain satisfied.*

Proof. A term-by-term audit of the CoreFalcon+ security reduction and security bound (in [Appendix A](#)) finds the only q dependent properties required for the reduction are the cyclotomic ring setting R_q and $s \geq \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \|B\|_{GS}$. We have already established in [Lemma 3.1](#) that the CoreFalcon+ parameterization results in $s \geq \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \|B\|_{GS}$ being independent of q . Therefore, so long as q permits the cyclotomic ring structure R_q , modification of q does not affect the applicability of the reduction. □

Corollary 3.3 (Modulus-independent normalized SIS target). *For $m = 2n$,*

$$\frac{\beta}{(\det \Lambda_q^\perp(A))^{1/m}} = \frac{\beta}{q^{n/m}} = \frac{\beta}{\sqrt{q}},$$

and therefore the target norm remains constant relative to the determinant scale of the q -ary lattice under a change of modulus.

Proof. The same argument holds for $\beta = \tau s \sqrt{2n}$ with respect to s . Notice $\beta/s = \tau \sqrt{2n}$ is also independent of q , obeying the same $\sqrt{q}$ scaling behavior. Increasing q raises both s and β proportionally, so β itself scales as $\sqrt{q}$, while the normalized quantity $\beta/\sqrt{q}$ is independent of q . The hardness of t - $\mathcal{R}$ -ISIS will depend on the shortness of ℓ_2 norm threshold β relative to the natural scale of the lattice. With $m = 2n$ the natural scale of a q -ary lattice is $(\det \Lambda_q^\perp(A))^{1/m} = q^{n/m} = q^{1/2}$. So

$$\frac{\beta}{(\det \Lambda_q^\perp(A))^{1/m}} = \frac{\beta}{q^{n/m}} = \frac{\beta}{\sqrt{q}}$$

is the same q -independent quantity as before, contextualizing why the normalized SIS target length is stable across the modulus change. The normalized $\beta/\sqrt{q}$ remains constant under variation of q . □

Lattice Reduction Attacks on q -ary Lattices Next, to support the findings from the lattice-estimator output, we argue that these parameters result in predictable behavior for the estimated hardness of t - $\mathcal{R}$ -ISIS by analyzing lattice reduction attacks on q -ary lattices.

Let $\Lambda_q^\perp(A)$ be an m -dimensional q -ary lattice, where q is prime, $A \in \mathbb{Z}_q^{n \times m}$ is a random matrix, with linearly independent rows and $m \geq n$. The number of elements in $\Lambda_q^\perp(A) \bmod q$ is q^{m-n} . The determinant is $\det(\Lambda_q^\perp(A)) = q^n$.

$$\Lambda_q^\perp(A) = \{y \in \mathbb{Z}^m : Ay = 0 \bmod q\}$$

By Definition 2.11 of [17] The volume of $\Lambda_q^\perp(A)$ is $\det(\Lambda_q^\perp(A)) = q^n$, and corresponds to the sparsity of lattice points in $\mathbb{R}^m$. By (Section 3, [18]), a heuristic estimate can be made for the first successive minimum of $\Lambda_q^\perp(A)$ as the smallest radius of an m dimensional ball with volume q^n

$$\lambda_1(\Lambda_q^\perp(A)) \approx q^{n/m} \sqrt{\frac{m}{2\pi e}}$$

An initial observation is that the determinant of a q -ary SIS lattice scales exactly as q^n , and the heuristic estimate from [18] of the first successive minimum λ_1 scales proportional to $q^{n/m}$. For $m = 2n$, the estimated shortest vector in the lattice Λ_h is approximately

$$\lambda_1(\Lambda_q^\perp(A)) \approx q^{n/m} \sqrt{\frac{m}{2\pi e}} = q^{n/2n} \sqrt{\frac{2n}{2\pi e}} = q^{1/2} \sqrt{\frac{n}{\pi e}} = \sqrt{\frac{qn}{\pi e}}$$

At the time of equation (1), Section 3 of [18] (2008), the best known lattice reduction algorithms for determining short vectors in a random m dimensional q -ary lattice $\Lambda_q^\perp(A)$ were capable of solving for lengths close to

$$\min\{q, (\det(\Lambda_q^\perp(A)))^{1/m} \cdot \delta^m\} = \min\{q, q^{n/m} \delta^m\}$$

Where δ depends on the algorithm and is observed by [14] to be typically $\delta \approx 1.013$ for fast algorithms, and as low as $\delta \approx 1.011$ for slower more precise algorithms. For our parameter choices

$$\min\{q, q^{n/m} \delta^m\} = \min\{q, q^{n/2n} \delta^{2n}\} = \min\{q, \sqrt{q} \cdot \delta^{2n}\}$$

Using the script `bkz_length.py` in the supporting code, $\min\{q, q^{n/m} \delta^m\}$ is used to compare the signature norm bound β against these estimates. For the $\lambda = 128, n = 512$ setting δ would need to be at most 1.00387782756709. For the $\lambda = 256, n = 1024$ setting δ would need to be at most 1.00211437276073. Meaning for these style lattice reduction attacks to be effective, δ needs to be below these values of δ for the respective parameter sets. Which is out of practical reach by the analysis of [14].

3.3 Security Estimates

Now that we have established the viability of the CoreFalcon+ reduction for our use case, we re-estimate an adversary's advantage $\text{Adv}_{q, \alpha, \beta, B}^{(Q_H+1)\text{-R-ISIS}}$ in solving the $(Q_H + 1)$ -R-ISIS problem under the new modulus. The CoreFalcon+ approach assumes that t -R-ISIS is as hard as SIS and estimates the resulting computational security using the `lattice-estimator` library [3]. We briefly review this methodology to clarify both the origin of the computational security term and the role played by the reduction to t -R-ISIS before presenting the concrete estimates for our parameter sets. It is explicitly assumed that t -R-ISIS is as hard as SIS, in part because best known attacks do not rely on the ring structure of R-ISIS. Estimates of SIS are made using the core-SVP methodology

from section 6 of [2], which considers dual and primal BKZ / sieve style attacks for predicting the concrete hardness of SVP style problems. They do not consider enumeration style attacks because they run in super-exponential time, whereas BKZ runs in exponential time. This is also the methodology behind the `SIS.estimate.rough()` function in `lattice-estimator`. The core-SVP hardness estimates are:

n	λ	q	s	β	SIS estimate
512	128	2130706433	69011.59	$2.4292 \cdot 10^6$	$2^{121.18}$
1024	256	2130706433	70115.84	$3.4904 \cdot 10^6$	$2^{279.15}$

Running `scripts/corefalcon.sage` from the supporting code evaluates the Theorem 1 UF-CMA bound from a Closer Look at Falcon for the KoalaBear instantiation.

n	λ	SIS bits	UF-CMA bits
512	128	121.18	117.98
1024	256	279.15	256.00

Rényi and sampler term decomposition from the same run:

n	λ	a_p	a_u	r_p	r_u	$p\text{PreSmp},\beta$
512	128	75.35	75.35	1.00	1.00	0.9999951
1024	256	150.70	150.70	1.00	1.00	0.999999998

3.4 Performance

Benchmarks for KoalaFalcon-512 signatures were collected on a MacBook Pro with the M4 Max chip with 48GB RAM. Throughput and latency for a run of 2^{17} signatures. Signatures are 2.1 KB. Multi-threaded benchmarks give each thread a single-threaded signing job, which independently run across 16 threads.

Mode	Threads	$\log_2 N$	Keygen (s)	Sign total (s)	Sign/s	Verify total (s)	Verify/s
Single-threaded	1	17	0.104	73.286	1788	9.566	13701
Multi-threaded	16	17	0.092	6.363	20599	0.816	160625

Table 2: Falcon-512 performance, 131,072 signatures.

Mode	Threads	$\log_2 N$	Keygen (s)	Sign total (s)	Sign/s	Verify total (s)	Verify/s
Single-threaded	1	17	0.795	166.747	786	36.205	3620
Multi-threaded	16	17	0.388	15.332	8549	3.421	38310

Table 3: Falcon-1024 performance, 131,072 signatures.

Runtime performance and signature size is strictly worse than the original Falcon scheme, but this is to be expected given the increase in field size (and therefore the data size of each ring coefficient).

Mode	Threads	$\log_2 N$	Keygen (s)	Sign total (s)	Sign/s	Verify total (s)	Verify/s
Single-threaded	1	17	0.010	13.601	9637	1.507	86999
Multi-threaded	16	17	0.002	1.162	112799	0.129	1017779

Table 4: FN-DSA / Falcon-512 performance, 131,072 signatures.

Mode	Threads	$\log_2 N$	Keygen (s)	Sign total (s)	Sign/s	Verify total (s)	Verify/s
Single-threaded	1	17	0.011	27.025	4850	2.980	43987
Multi-threaded	16	17	0.013	2.140	61255	0.247	531447

Table 5: FN-DSA / Falcon-1024 performance, 131,072 signatures.

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A Proofs of KoalaFalcon Requirements

A.1 Algebraic Feasibility Proofs

Proof 1: Ring Algebraic Requirements Let $q = 2^{31} - 2^{24} + 1$, a known prime. The field $\mathbb{F}_q$ has multiplicative group $\mathbb{F}_q^\times$ of cyclic order $q - 1$. There exists a primitive $2n$ -th root of unity in

$\mathbb{F}_q$ if and only if $2n|q-1$.

$$q-1 = 2^{31} - 2^{24} = 2^{24}(2^7 - 1)$$

For $n = 512 = 2^9$, $2^{24}(2^7 - 1) \equiv 0 \pmod{2^{10}}$. For $n = 1024 = 2^{10}$, $2^{24}(2^7 - 1) \equiv 0 \pmod{2^{11}}$. Therefore there exists a primitive $2n$ -th root of unity in $\mathbb{F}_q$ for both choices of n . The existence of a primitive $2n$ -th root of unity implies that $X^n + 1$ has n roots, splitting into n linear factors. ■

Proof 2: Public, Private Key Remain Bases to the Same Lattice Let $\Lambda = \Lambda(B_1)$ be a $2n$ -dimensional lattice with basis B_1 . B_2 is also a basis of Λ if and only if there exists a unimodular matrix U with determinant ± 1 such that $UB_2 = B_1$. This definition is adapted from Theorem 2.9 of [17].

Let $q = 2^{31} - 2^{24} + 1$, $n = 2^\kappa$ is a power-of-two, $R = \mathbb{Z}[X]/(X^n + 1)$, with $\phi = X^n + 1$ fully splits into n linear factors over $\mathbb{Z}_q$, and $f, g \in R_q^\times$. The relation

$$\begin{aligned} h &\equiv gf^{-1} \pmod{\phi, q}, \\ fG - gF &\equiv q \pmod{\phi} \end{aligned}$$

The private basis B_1 and public basis B_2 of a $2n$ dimensional NTRU lattice are defined below.

$$B_1 = \begin{bmatrix} g & -f \\ G & -F \end{bmatrix}, \quad B_2 = \begin{bmatrix} -h & I_n \\ qI_n & O_n \end{bmatrix}.$$

We wish to show there exists a unimodular matrix U with integral coefficients such that $UB_2 = B_1$. One such matrix is

$$U = \begin{bmatrix} -f & \frac{g-hf}{q} \\ -F & \frac{G-hF}{q} \end{bmatrix}.$$

Evaluating UB_2 gives

$$\begin{aligned} UB_2 &= \begin{bmatrix} -f & \frac{g-hf}{q} \\ -F & \frac{G-hF}{q} \end{bmatrix} \begin{bmatrix} -h & I_n \\ qI_n & O_n \end{bmatrix} \\ &= \begin{bmatrix} -f(-h) + \frac{g-hf}{q} qI_n & (-f)I_n + \frac{g-hf}{q} O_n \\ -F(-h) + \frac{G-hF}{q} qI_n & -F I_n + \frac{G-hF}{q} O_n \end{bmatrix} \\ &= \begin{bmatrix} fh + g - hf & -f \\ Fh + (G - hF) & -F \end{bmatrix} \\ &= \begin{bmatrix} g & -f \\ G & -F \end{bmatrix} = B_1. \end{aligned}$$

The determinant of U is

$$\begin{aligned} \det(U) &= (-f \cdot \frac{G-hF}{q}) - (-F \cdot \frac{g-hf}{q}) \\ &= \frac{-fG + fhF}{q} - \frac{-Fg + Fhf}{q} = \frac{-fG + fhF + Fg - Fhf}{q} = \frac{-fG + Fg}{q} \end{aligned}$$

And since $fG - gF \equiv q \pmod{X^n + 1}$,

$$\det(U) = \frac{-fG + Fg}{q} = -\frac{q}{q} = -1$$

Therefore there exists a unimodular matrix U with integral coefficients such that $UB_2 = B_1$. Thus B_1, B_2 are bases of the same lattice Λ . ■

Proof 3: Standard Deviation is Greater than Smoothing Parameter The loss-term corollaries require $s \geq \eta_\epsilon(\Lambda_{h,q})$. By the GPV smoothing lemma (Lemma 3.1 of [15]), for the trapdoor basis $B_{f,g}$ produced by TpdGen,

$$\eta_\epsilon(\Lambda_{h,q}) \leq \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \|B_{f,g}\|_{GS}.$$

By Lemma 4 of [12], $\eta_\epsilon(\mathbb{Z}^{2n}) \leq \eta_\epsilon^{\text{upper}}(\mathbb{Z}^{2n})$. With $\|B_{f,g}\|_{GS} \leq \alpha\sqrt{q}$ and the CoreFalcon+ choice of s ,

$$\eta_\epsilon(\Lambda_{h,q}) \leq \eta_\epsilon^{\text{upper}}(\mathbb{Z}^{2n}) \cdot \alpha\sqrt{q} = s.$$

■

For intuitive context, the trapdoor-quality bound $\|B_{f,g}\|_{GS} \leq \alpha\sqrt{q}$ forces the relevant smoothing threshold to scale as $\sqrt{q}$, and the definition of s matches it. Consider the integer lattice $\mathbb{Z}^{2n}$ scaled by $\|B_{f,g}\|_{GS}$. Since $\Lambda_{h,q}$ may be skewed, the scaled integer lattice $\|B_{f,g}\|_{GS} \cdot \mathbb{Z}^{2n}$ represents the case where every orthogonal direction of a $2n$ -dimensional lattice is of length equal to the worst-case Gram-Schmidt norm of $\|B_{f,g}\|_{GS}$. Smoothing this scaled lattice requires width at least $\|B_{f,g}\|_{GS} \cdot \eta_\epsilon(\mathbb{Z}^{2n})$, which is exactly the CoreFalcon+ choice of s up to the Lemma 4 upper bound. Raising q raises both sides by the same $\sqrt{q}$ factor, so the condition is preserved.

B q -Dependence Audit of CoreFalcon+ Security Bound

This section re-derives the security bound for CoreFalcon+, examining term-by-term where exactly the modulus plays a role in their argument. The derivations in this audit originate from the CoreFalcon+ paper.

r_u : random-oracle / syndrome uniformity r_u controls the Rényi divergence between the uniform distribution $U(R_q)$ and the distribution of the programmed syndrome $s_1 + s_2 h \bmod q$ for Gaussian $s_1, s_2 \sim D_{R,s}$:

$$r_u = \max_{(\cdot, h) \in \text{TpdGen}} R_{a_u}(P \| Q_h), \quad P = U(R_q).$$

Where Q_h is the distribution of $s_1 + s_2 \cdot h \bmod q$ with $s_1, s_2 \sim D_{R,s}$ and signature standard deviation

$$s = \frac{1}{\pi} \sqrt{\frac{\ln\left(4n\left(1 + \frac{1}{\epsilon}\right)\right)}{2}} \cdot \alpha\sqrt{q}.$$

By Corollary 2.11 [12] (proof in Appendix B.3 of [12]), with $P = U(R_q)$ and Q the distribution of $u + v \cdot h \bmod q$ for $u, v \sim D_{R,s}$, the syndrome-uniformity Rényi divergence $R_a(P \| Q)$ obeys the bound stated there. The only nontrivial pre-condition behind it is $s \geq \eta_\epsilon(\Lambda_{h,q})$. Bases B_h and $B_{f,g}$ of $\Lambda_{h,q}$ are defined by $\text{TpdGen}(\alpha, q) \rightarrow (f, g, F, G, h)$ where $h \in R_q \setminus \{0\}$ and $(f, g, F, G) \in R^4$. The trapdoor basis satisfies $\|B_{f,g}\|_{GS} \leq \alpha\sqrt{q}$. Proof 3 then resolves this condition, confirming

$$r_u \lesssim 1 + \frac{2a_u \epsilon^2}{(1 - \epsilon)^2}.$$

Once the smoothing condition holds, this bound has no remaining q -dependence beyond the cyclotomic ring setting and $\|B_{f,g}\|_{GS} \leq \alpha\sqrt{q}$ for Proof 3 to apply.

r_p : sampler output versus ideal Gaussian r_p is the Rényi divergence between the pre-image sampler output $\text{PreSmp}(B, s, (c, 0))$ and the ideal discrete Gaussian $D_{\Lambda, s}$

$$r_p = \max_{(B, \cdot) \in \text{TpGen}, c \in R_q} R_{ap}(\text{PreSmp}(B, s, (c, 0)) \parallel D_{\Lambda, s}),$$

where $\Lambda = \Lambda(B)_{(c, 0)}$ is the coset obtained by shifting the lattice by the syndrome $(c, 0)$. By [Corollary 2.12 \[12\]](#), for $s \geq \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \|B\|_{GS}$ and arbitrary syndrome $c \in R_q$, the Rényi divergence is $R_a(\text{PreSmp}(B, s, (c, 0)) \parallel D_{\Lambda, s}) \lesssim 1 + 2a\epsilon^2$. This pre-condition is demonstrated to be the case in Proof 3 for the CoreFalcon+ parameter formula of s , therefore $r_p \lesssim 1 + 2a_p\epsilon^2$ applies in our case. Beyond this, [Corollary 2.12](#) makes no reference to q , depending only on n , ϵ , and $\|B\|_{GS}$. Though it remains a concern whether the actual sampler implementation still behaves as expected with 53-bit floating point arithmetic with the larger 31-bit modulus.

$p_{\text{PreSmp}, \beta}$: sampler success / rejection probability $p_{\text{PreSmp}, \beta}$ is the worst-case probability of sampling $(s_1, s_2) \in R^2$ with $\|(s_1, s_2)\|_2 \leq \beta$ from the FFO sampler. The derivation below is found in appendix E1 of [12]. We walk through it here to understand if at any point we invoke some property of q . At a high level it proceeds in four steps:

1. Compare PreSmp to $D_{\Lambda, s}$ using the relative-error bound ([Lemma 2.13 \[12\]](#)).
2. Apply the Gaussian tail bound to $D_{\Lambda+t, s}$ ([Lemma 2.15 \[12\]](#)).
3. Use smoothing ([Lemma 2.16 \[12\]](#)) to bound $\rho_s(\Lambda)/\rho_s(\Lambda + t)$.
4. Substitute $\beta = \tau s\sqrt{2n}$, causing the s -scale, and hence the q -scale, to cancel.

Let

$$p_{\text{PreSmp}, \beta} := \min_{(B, \cdot) \in \text{TpGen}, c \in R_q} \Pr_{(s_1, s_2) \leftarrow \text{PreSmp}(B, s, (c, 0))} [\|(s_1, s_2)\|_2 \leq \beta]$$

Take the complement

$$p_{\text{PreSmp}, \beta} := \min_{(B, \cdot) \in \text{TpGen}, c \in R_q} 1 - \Pr_{(s_1, s_2) \leftarrow \text{PreSmp}(B, s, (c, 0))} [\|(s_1, s_2)\|_2 > \beta]$$

By [Lemma 2.13 \[12\]](#), once $s \geq \eta_\epsilon(\mathbb{Z}^{2n}) \cdot \|B\|_{GS}$ the relative error $\delta_{\text{RE}}(\text{PreSmp}(B, s, (c, 0)), D_{\Lambda, s})$ is bounded by approximately 2ϵ .

This relative error quantifies the error in treating the discrete Gaussian $D_{\Lambda, s}$ as the same distribution as the distribution of the outputs of $\text{PreSmp}(B, s, (c, 0))$ over the lattice Λ , and is bounded by approximately 2ϵ . That is to say, we can let $D_{\Lambda, s}$ stand in for the distribution of $\text{PreSmp}(B, s, (c, 0))$ up to a relative error bounded by 2ϵ .

[Lemma 2.14 \[12\]](#) transfers a relative-error bound $\delta_{\text{RE}}(P, Q) = \delta$ between two distributions on $\mathbb{Z}^n$ into the tail bound $\Pr_{x \leftarrow P}[\|x\|_2 > \beta] \leq \Pr_{x \leftarrow Q}[\|x\|_2 > \beta] (1 + \delta)$ for any $\beta \geq 0$.

Then, by application of [Lemma 2.13](#) to [Lemma 2.14](#), we have for

$$\delta_{\text{RE}}(\text{PreSmp}(B, s, (c, 0)), D_{\Lambda(B)+t, s}) \lesssim 2\epsilon$$

$$\Pr_{(s_1, s_2) \leftarrow \text{PreSmp}(B, s, (c, 0))} [\| (s_1, s_2) \|_2 > \beta] \leq \Pr_{(s_1, s_2) \leftarrow D_{\Lambda(B)+t, s}} [\| (s_1, s_2) \|_2 > \beta] \cdot (1 + 2\epsilon), \quad t \in \Lambda(B)_{(c, 0)}.$$

Application to our current bound gives

$$\begin{aligned} & \min_{(B, \cdot) \in \text{TpGen}, c \in R_q} 1 - \Pr_{(s_1, s_2) \leftarrow \text{PreSmp}(B, s, (c, 0))} [\| (s_1, s_2) \|_2 > \beta] \\ & \gtrsim \min_{(B, \cdot) \in \text{TpGen}, c \in R_q} \left(1 - \Pr_{(s_1, s_2) \leftarrow D_{\Lambda(B)_{(c, 0)}, s}} [\| (s_1, s_2) \|_2 > \beta] \cdot (1 + 2\epsilon) \right) \end{aligned}$$

By [Lemma 2.15 \[12\]](#), the discrete Gaussian $D_{\Lambda+t, s}$ satisfies the tail bound $\Pr_{z \leftarrow D_{\Lambda+t, s}} [\|z\|_2 > \tau s \sqrt{n}] \leq \frac{\rho_s(\Lambda)}{\rho_s(\Lambda+t)} (\sqrt{e^{1-\tau^2}} \tau^2)^n$.

Applying to our current bound, and since the parameter $\beta = \tau s \sqrt{2n}$, we have

$$\begin{aligned} & \min_{(B, \cdot) \in \text{TpGen}, c \in R_q} \left(1 - \Pr_{(s_1, s_2) \leftarrow D_{\Lambda(B)_{(c, 0)}, s}} [\| (s_1, s_2) \|_2 > \beta] \cdot (1 + 2\epsilon) \right) \\ & \gtrsim \min_{(B, \cdot) \in \text{TpGen}, c \in R_q} \left(1 - \frac{\rho_s(\Lambda(B))}{\rho_s(\Lambda(B) + t)} \cdot \left(\sqrt{e^{1-\tau^2}} \tau^2 \right)^{2n} \cdot (1 + 2\epsilon) \right), \quad t \in \Lambda(B)_{(c, 0)}. \end{aligned}$$

By [Lemma 2.16 \[12\]](#), for $s \geq \eta_\epsilon(\Lambda)$ the Gaussian mass satisfies $\rho_{s, c}(\Lambda) \in [1 - \epsilon, 1 + \epsilon] \cdot \frac{(\sqrt{2\pi} \cdot s)^n}{\det(\Lambda)}$.

Since $\Lambda(B) + t$ is a translate (coset) of $\Lambda(B)$, the covolume is unchanged by translation, so $\det(\Lambda(B) + t) = \det(\Lambda(B))$, and $s \geq \eta_\epsilon(\Lambda(B))$ by [Proof 3](#). This gives

$$p_{\text{PreSmp}, \beta} \geq 1 - \left(\frac{1 + \epsilon}{1 - \epsilon} \cdot \left(\sqrt{e^{1-\tau^2}} \tau^2 \right)^{2n} \cdot (1 + 2\epsilon) \right).$$

This lower bound depends only on n , τ , and ϵ , not on q . Since β is always $\tau s \sqrt{2n}$, increasing q increases both s and β proportionally, and the s -scale cancels at the [Lemma 2.15](#) step.

Optimal Rényi orders a_u, a_p, C_s All choices $a_u, a_p \in \mathbb{R}^{\geq 1}, C_s \in \mathbb{N}^{\geq 1}$ are by definition proof constants and do not depend on q .

The Rényi orders a_u and a_p are downstream of the bounds already established for r_u and r_p . Once r_u and r_p are bounded by expressions depending only on ϵ and the chosen orders, optimizing the orders to minimize the overall loss introduces no new direct dependence on q . [Appendix E2 of \[12\]](#) reports the optimal a_p as

$$a_p = \frac{\lambda \epsilon^2 \ln(4) + \sqrt{8C_s \lambda \epsilon^2 \ln(2) + \lambda^2 \epsilon^4 \ln^2(4)}}{4C_s \epsilon^2}$$

The optimal a_u is determined analogously from the r_u bound. Conceptually, both a_u and a_p depend on λ , ϵ , C_s , and the Rényi-loss optimization, not directly on q . Numerical values are reported under [Concrete Parameter Choices](#).